



Artificial Intelligence Lab (AI-2002)

Lab Task 9

Task 1: Bayesian Network for Job Hiring

You are modeling whether a candidate gets hired.

Nodes:

- Education (E) — {High, Low}
- Experience (X) — {Experienced, Inexperienced}
- Interview (I) — {Good, Bad}
- HiringDecision (H) — {Hired, NotHired}

Dependencies:

- Education and Experience influence Interview performance
- Interview influences HiringDecision

Prior Probabilities:

- $P(\text{Education} = \text{High}) = 0.65$, $\text{Low} = 0.35$
- $P(\text{Experience} = \text{Experienced}) = 0.5$, $\text{Inexperienced} = 0.5$

Tasks to do:

- Draw the Bayesian Network structure
- Define **CPTs** for:
 - Interview | Education, Experience
 - HiringDecision | Interview
- Implement the network using Python (pgmpy or similar)



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Task 2: Bayesian Network for Car Fault Diagnosis

You are diagnosing whether a car has an issue based on symptoms.

Nodes:

- Fault (F) — {EngineIssue, BatteryIssue}
- Symptoms:
 - CarWon'tStart (S) — {Yes, No}
 - DimLights (D) — {Yes, No}
 - StrangeNoise (N) — {Yes, No}

Network Structure:

- Fault $\rightarrow$ CarWon'tStart
- Fault $\rightarrow$ DimLights
- Fault $\rightarrow$ StrangeNoise

Prior Probabilities:

- $P(\text{EngineIssue}) = 0.4$
- $P(\text{BatteryIssue}) = 0.6$

Conditional Probabilities (example assumptions):

- $P(\text{CarWon'tStart} = \text{Yes} \mid \text{EngineIssue}) = 0.85$
- $P(\text{CarWon'tStart} = \text{Yes} \mid \text{BatteryIssue}) = 0.7$
- $P(\text{DimLights} = \text{Yes} \mid \text{EngineIssue}) = 0.3$
- $P(\text{DimLights} = \text{Yes} \mid \text{BatteryIssue}) = 0.8$
- $P(\text{StrangeNoise} = \text{Yes} \mid \text{EngineIssue}) = 0.75$
- $P(\text{StrangeNoise} = \text{Yes} \mid \text{BatteryIssue}) = 0.2$

Tasks to do:

1. Construct the Bayesian Network diagram
2. Define full CPTs for all nodes
3. Perform inference:



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Task 3: Bayesian Network for Online Shopping Behavior

You are modeling whether a user makes a purchase.

Nodes:

- AdExposure (A) — {Yes, No}
- WebsiteExperience (W) — {Good, Poor}
- ProductPrice (P) — {High, Low}
- Purchase (C) — {Yes, No}

Dependencies:

- AdExposure influences WebsiteExperience
- WebsiteExperience and ProductPrice influence Purchase

Prior Probabilities:

- $P(\text{AdExposure} = \text{Yes}) = 0.6$
- $P(\text{WebsiteExperience} = \text{Good} \mid \text{AdExposure} = \text{Yes}) = 0.8$
- $P(\text{WebsiteExperience} = \text{Good} \mid \text{AdExposure} = \text{No}) = 0.4$
- $P(\text{ProductPrice} = \text{Low}) = 0.55$
- $P(\text{ProductPrice} = \text{High}) = 0.45$

Tasks to do:

- Draw the Bayesian Network structure
- Define full CPTs for Purchase
- Implement using Python (pgmpy or similar)